

MedHop-Agent: Efficient Agentic Multi-Hop Retrieval for Biomedical Question Answering

Team Members: Carol Li, Selina Qian, Behjat Riyaz, Ken Su

Problems and Motivation

Biomedical question answering often requires combining evidence across multiple documents and reasoning steps. MedQA tests whether models can apply broad clinical knowledge to USMLE-style scenarios, while MedHop requires linking evidence across biomedical entities and documents to infer drug-drug interactions [5, 19]. Retrieval-augmented generation (RAG) improves factual grounding by retrieving external evidence, but a standard single-shot top- k retrieval pipeline is brittle on multi-hop questions because later evidence needs depend on earlier partial inferences [9, 18, 8]. When the retrieved context is incomplete or poorly ordered, downstream reasoning can fragment and factual errors can compound, which is especially problematic in biomedical settings where precision matters [10]. Improving biomedical retrieval and explicit multi-step reasoning is therefore central to building more reliable clinical question-answering systems.

Baselines and Current SOTA Approaches

Classical retrieval baselines for question answering include sparse ranking with BM25 and dense retrieval with dual-encoder or sentence-embedding models such as SBERT and DPR [16, 15, 7]. A common baseline is vanilla RAG, in which the system retrieves relevant passages once and conditions generation on that fixed evidence set [9]. Recent state-of-the-art approaches extend this paradigm in two main directions. Graph-based systems such as GRAG, QCG-RAG, and StepChain GraphRAG organize evidence as connected subgraphs to support structured multi-hop retrieval and more interpretable evidence chains [1, 20, 13]. In parallel, agentic RAG systems let the model plan retrieval actions, choose among tools, and iteratively refine context, improving flexibility on complex queries but often increasing system and inference cost [17, 18, 4].

Proposed Extension(s)

We propose an efficiency-aware agentic multi-hop RAG pipeline for biomedical QA with three trainable components. First, we initialize the retriever with MedCPT and fine-tune it on question-passage triples from MedHop and MedHopQA using a contrastive loss with in-batch negatives and hard negatives mined from BM25 and the current dense retriever, so the model learns to separate truly supporting biomedical passages from lexically similar distractors [6, 7, 16]. Second, we train a lightweight reranker on the same candidate pool using a pairwise ranking objective, promoting gold evidence above near-miss passages; for efficiency, only the top 20–30 retrieved passages are reranked

and only the top 5–8 are passed to the generator. For MedQA, where gold supporting passages are less explicit, we will use weak supervision by treating passages retrieved for the correct answer choice with high biomedical entity overlap as pseudo-positives and passages linked to distractor options as pseudo-negatives.

The controller/router will be a small 3-way classifier that predicts whether a question should use single-pass retrieval, 2-hop retrieval, or 3-hop retrieval. Its input features will combine a cheap question embedding with low-cost retrieval signals from the first pass, including query length, number of recognized biomedical entities, BM25–dense score gap, top- k score entropy, and agreement between sparse and dense retrieval. We will generate router labels automatically on the training split: if the answer or supporting evidence is recovered with one retrieval pass, the instance is labeled easy; if iterative retrieval improves evidence recall or answer accuracy, it is labeled multi-hop, following the adaptive-routing intuition of Adaptive-RAG [4]. At inference time, easy questions follow vanilla RAG, while harder ones use IRCOT-style retrieve–reason cycles with condensed memory and early stopping once reranker confidence stops improving, which keeps the agentic design concrete, trainable, and compute-aware [18, 8].

Datasets and Evaluation Metrics

To evaluate the proposed framework, we use three biomedical QA benchmarks. MedQA is a collection of USMLE-style multiple-choice questions that tests broad clinical reasoning [5]. MedHop requires models to connect evidence across documents about drugs and proteins in order to answer multi-hop biomedical queries [19]. As a newer benchmark, MedHopQA from the BioCreative IX shared task provides 1,000 disease-, gene-, and chemical-focused multi-hop questions; the task uses short answers and optional long answers to evaluate both answer precision and reasoning quality [2]. We report Recall@5, nDCG@ K , and MAP for retrieval quality, and Exact Match for answer correctness [11, 3, 14]. For MedHopQA, we additionally use the shared task’s concept-level score to measure semantic correctness beyond exact string matching [2, 12].

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